

★ Determinant ★ [Chap - 12]

★ Sq³ Array $\rightarrow$ determinant.

★ Minor & Co-factors.

★ Sarrus Rule



$$\Rightarrow \Delta = \{a_n z + b q x + c m y\} - \{x n c + y p c + z m b\}$$

★ Properties of Determinant ★

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

[a] $\Delta = \Delta^T$

[b] m multiply any Row/column $\Delta^{\text{New}} = m \Delta$

[c] $R_i \leftrightarrow R_j$ OR $C_i \leftrightarrow C_j$ $\Delta^{\text{New}} = -\Delta$, n = odd
 $\Delta^{\text{New}} = \Delta$, n = Even.

[d] Sum/Diff.

$$\Delta = \begin{vmatrix} a_1 + d_1 & a_2 + d_2 & a_3 + d_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \Rightarrow \begin{vmatrix} a_1 & d_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \pm \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

[e] Sigma of Determinant.

$$\Delta x = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \Rightarrow \Sigma \Delta x = \begin{vmatrix} \Sigma f_1(x) & \Sigma f_2(x) & \Sigma f_3(x) \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

[f] Differentiation / Integration.

$$\Delta \Rightarrow \Delta'(\text{diff}) = \begin{vmatrix} \text{1st} \\ \text{Row/Col} \end{vmatrix} + \begin{vmatrix} \text{2nd} \\ \text{Row/Col} \end{vmatrix} + \begin{vmatrix} \text{3rd} \\ \text{Row/Col} \end{vmatrix}$$

$$\Delta \Rightarrow \int \Delta = \begin{vmatrix} \int f_1(x) & \int f_2(x) & \int f_3(x) \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

[h] $R_i \rightarrow R_i + m R_j$ OR $C_i \rightarrow C_i + m C_j$ } New determinant but value does not change.

★ Some determinant MUST be learn ★

$$[a] \Delta = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

$$[b] \Delta = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x+y+z)(x-y)(y-z)(z-x)$$

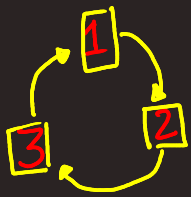
$$[c] \Delta = \begin{vmatrix} 1 & 1 & 1 \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = (xy+yz+zx)(x-y)(y-z)(z-x)$$

$$[d] \Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -[a+b+c][a^2+b^2+c^2-ab-bc-ca]$$

★ Application of Determinant ★

[a] Area of Δ to given vertices ★

$$(x_1, y_1), (x_2, y_2), (x_3, y_3) \quad A = \frac{1}{2} \left| x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right|$$



$$A = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

[B] Concurrent lines

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

$$[C] \quad ax^2 + by^2 + 2hxy + 2fy + 2gx + c \equiv 0$$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

★ Prod of two determinant ★

$$\Delta = \Delta_1 \times \Delta_2 \rightarrow \text{it would be}$$

$$\left. \begin{matrix} R \times R \\ C \times C \\ R \times C \\ C \times R \end{matrix} \right\}$$

★ Determinant form by co-factor, $\Delta^{\text{new}} = [\Delta^{\text{old}}]^{n-1}$, $n = \text{order}$.

★ System of linear eqn system ★

[A] Heterogeneous

- [a] 2 line in 2 variable } line
- [b] 3 line in 2 variable }
- [c] 3 line in 3 variable } Plane

[B] Homogeneous

[a] 3 lines in 3 Variable

[A] 2 lines in 2 variable.

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$\Delta_1 = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}$$

for Unique/intersection $\left\{ x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta} \right\}$

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \quad y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

[ii] parallel line. $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

[iii] Coincide line $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

★ 3 lines in 2 Variable (concurrent)

★ 3 lines in 3 variables ★

C ①

Consistent

Infinite

$$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$$

[A] ①

Unique
Consistent
a solution

[B] ③

Non-consistent
No-solution

$$\Delta = 0$$

but $\Delta_1, \Delta_2, \Delta_3$
are not = 0 at
same time

$$a_1x + b_1y + c_1z + d_1 = 0$$

$$a_2x + b_2y + c_2z + d_2 = 0$$

$$a_3x + b_3y + c_3z + d_3 = 0$$

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta_1 = - \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} c_1 & d_1 & a_1 \\ c_2 & d_2 & a_2 \\ c_3 & d_3 & a_3 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} d_1 & a_1 & b_1 \\ d_2 & a_2 & b_2 \\ d_3 & a_3 & b_3 \end{vmatrix}$$

$$x = \frac{\Delta_1}{\Delta}$$

$$y = \frac{\Delta_2}{\Delta}$$

$$z = \frac{\Delta_3}{\Delta}$$

No - Camer

$$\text{No-soln} \left\{ \frac{1 \& 2}{3} = \dots = \dots \neq \dots \right.$$

{ Infinite soln }

$$\frac{1 \& 2}{3} = \frac{1 \& 2}{3} = \dots = \dots$$

[B] Homogeneous *
(0,0,0) trivial soln. } single soln.

Infinite soln / Non-trivial soln

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0.$$

$$PYQ = \{4, 9, 10, 12, 18, 25, 30, 31, 37, \textcircled{39}, 45, 57, 58, 65, 66, 68, 79, 80, 81, 84, 86, 91, 97\}_{100}.$$

★ MATRICES ★ Chapter. 13

- Def = arrⁿ of array in Rectangular form $\{m \times n\}$ order of Matrix. → Row
↓ Column.
 - Classification of Matrices / Types =
 - [a] Row
 - [b] column
 - [c] Null
 - [d] Diagonal
 - [e] scalar
 - [f] Unit / Identity
 - [g] Triangular Matrix
 - [h] Symmetric Matrix
 - [i] skew-symmetric matrix
 - [k] Determinant of Matrix [Square]
- Square $\Delta = 0$ Singular $\Delta \neq 0$ Non-singular
- [l] Trace of Matrix
 - [m] multiply by m →
 - [n] Addition & Subtraction of two Matrices.
 - [o] Equal Matrices.

[B] Product of two matrices *

$$M_{m \times n} \times N_{n \times y} \rightarrow P_{m \times y}$$

$$n = n$$

$$A \rightarrow \text{Symmetric} \rightarrow A^h = \text{Symmetric}$$

$$A \rightarrow \text{Skew-symme} \rightarrow A^{\text{even}} = \text{Symmetric}$$

$$A^{\text{odd}} = \text{skew symmetric}$$

$$A = [a_1, a_2, a_3, a_4, a_5, \dots, a_n] \text{ diago}$$

$$B = [b_1, b_2, b_3, \dots, b_n]$$

$$A \times B = [a_1 b_1, a_2 b_2, a_3 b_3, \dots, a_n b_n]$$

$$A = [a_1, a_2, a_3, a_4, \dots, a_n]$$

$$A^n = [a_1^n, a_2^n, a_3^n, a_4^n, \dots, a_n^n]$$

★ A-Square Matrix, $f(A) = a_0 I + a_1 A + a_2 A^2 + a_3 A^3 + a_4 A^4 + \dots + a_n A^n$
 $a_0, a_1, a_2, \dots, a_n$ scalar quantity chain.

★ Special Matrices ★

[a] Orthogonal Matrix:

$$AA^T = A^T A = I, \quad \det(A) = \pm 1$$

[b] Idempotent Matrix:

$$A^2 = A, \quad \det(A) = 0, 1.$$

[c] Involuntary Matrix:

$$A^2 = I, \quad \det(A) = \pm 1$$

[d] Nilpotent Matrix:

$$A^m = 0, \quad \det(A) = 0$$

★ Properties of Matrices ★

[a] $AB = BA$ Not necessarily true.

$$\nabla [A+B]^2 \neq A^2 + 2AB + B^2$$

$$[A+B]^2 = A^2 + AB + BA + B^2$$

★ if $AB = BA$ then $[A+B]^2 = A^2 + 2AB + B^2$.

$$\underline{\underline{\text{So}}} (A+B)^n = {}^nC_0 A^n B^0 + {}^nC_1 A^{n-1} B^1 + \dots + {}^nC_n A^0 B^n$$

$$[b] A(BC) = (AB)C$$

$$[c] A(B \pm C) = AB \pm AC$$

[d] if $AB = 0$ Not necessary at least A or B should be zero matrix.

$$[e] AB = AC \longrightarrow \nabla B = C$$

$$[f] \text{tr}(AB) = \text{tr}(BA)$$

$$[g] (AB)^T = B^T A^T$$

$$[h] \det(ABCD) = \det(A) \cdot \det(B) \cdot \det(C) \cdot \det(D).$$

[C] Adjoint of Matrix: Co-factor nikalo and Use transpose kr do.

{adj(A).}

Properties :

$$[a] A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = |A| I_n$$

$$|\text{adj}(A)| = |A|^{n-1}$$

$$[b] |\text{Adj} \cdot \text{adj}(A)| = |A|^{(n-1)^2}$$

$$|\text{Adj} \cdot \text{adj} \cdot \text{adj} \cdot \text{adj}(A)| = |A|^{(n-1)^4}$$

$$[c] \text{adj} \cdot \text{adj}(A) = |A|^{n-2} \cdot A$$

$$[d] \text{Adj}(A \cdot B) = \text{adj}(B) \cdot \text{adj}(A)$$

$$[e] \text{Adj}(A^T) = [\text{Adj} A]^T$$

$$[f] \text{adj}(KA) = K^{n-1} \text{adj}(A)$$

[D] Invers of Matrix.

$$\text{if } |A| \neq 0, \quad A A^{-1} = I$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

Properties :

$$[b] (A^{-1})^{-1} = A.$$

$$[c] |A^{-1}| = |A|^{-1} = \frac{1}{|A|}$$

$$[d] (AB)^{-1} = B^{-1} \cdot A^{-1}$$

$$(ABCD)^{-1} = D^{-1} \cdot C^{-1} \cdot B^{-1} \cdot A^{-1}$$

$$[e] (A^{-1})^T = (A^T)^{-1}$$

$$[f] (A^k)^{-1} = (A^{-1})^k$$

$$(g) (kA)^{-1} = \frac{A^{-1}}{k}$$

$$(h) \text{Adj}(A^{-1}) = \frac{A}{|A|}$$

$$[i] A = \text{diag} [a_1, a_2, a_3, a_4, \dots, a_n]$$

$$A^{-1} = \text{diag} [a_1^{-1}, a_2^{-1}, a_3^{-1}, a_4^{-1}, \dots, a_n^{-1}]$$

[E] Characteristics Roots of Matrix.

$$\left\{ \begin{array}{l} A^2 - \text{trace}(A) \cdot A + |A| = 0 \rightarrow \text{order } 2 \times 2 \\ A^3 - \text{trace}(A) \cdot A^2 + \underbrace{\text{trace}(A^2)}_{\text{Minor of Diagonal element}} A - |A| = 0 \rightarrow \text{order } 3 \times 3 \end{array} \right.$$

★ Complex No ★ Chapter (3)

[A] Cube root of unity. $x^3 = 1$

$$\begin{array}{c} \nearrow 1 \\ \rightarrow \frac{-1 + \sqrt{3}i}{2} \quad w \\ \searrow \frac{-1 - \sqrt{3}i}{2} \quad w^2 \end{array}$$

$$x^3 = -1 \begin{array}{c} \nearrow -1 \\ \rightarrow -w \\ \searrow -w^2 \end{array}$$

* Properties: Each Complex Root is square of other. like $w^2 \rightarrow w^2$
 $[w^2]^2 \rightarrow w$

ii] Integral Power of w

$$\begin{array}{l} w^{3h} \rightarrow 1 \\ w^{3h+1} \rightarrow w \\ w^{3h+2} \rightarrow w^2 \end{array}$$

$$[iii] x^3 = -1 \begin{array}{c} \nearrow -1 \\ \rightarrow -w \\ \searrow -w^2 \end{array}$$

$$x^3 = 27 \begin{array}{c} \nearrow 3 \\ \rightarrow 3w \\ \searrow 3w^2 \end{array}$$

$$x^3 = a^3 \begin{array}{c} \nearrow a \\ \rightarrow aw \\ \searrow aw^2 \end{array}$$

[iv] $1 + w + w^2 = 0$, $1 \times w \times w^2 = 1$, w Reciprocal $\frac{1}{w^2}$

$$w^2 = \frac{1}{w}$$

[v] $1 + w^n + w^{2n} = \begin{cases} 0 & n \neq \text{Mof } 3 \\ 3 & n = \text{Mof } 3 \end{cases}$

[vi] $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

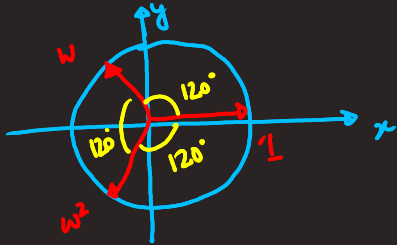
$$(a+b)(a+bw)(a+bw^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$[a-b][a-bw][a-bw^2]$$

[vii] $a^3 + b^3 + c^3 - 3abc = [a+b+c][a^2 + b^2 + c^2 - ab - bc - ca]$

$$[a+b+c][a+bw+cw^2][a+bw^2+cw]$$



★ n^{th} Root of '1' and '-1'

$$x^3 = 1, +1, w, w^2 \left\{ e^{i \frac{2k\pi}{n}}, e^{i \frac{4k\pi}{n}} \right\}$$

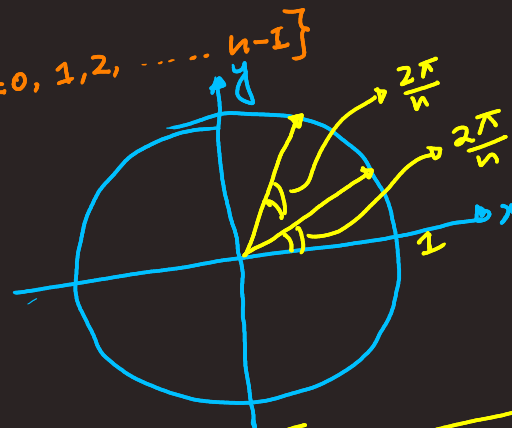
$$\left\{ 0 + i0, \left(\frac{2\pi}{3} + i \frac{2\pi}{3} \right), \left(\frac{4\pi}{3} + i \frac{4\pi}{3} \right) \right\}$$

★ $x^n = 1$ $n = \text{odd}$
 $1 \rightarrow \text{Real}(1)$
 $(n-1) \rightarrow \text{con \& conjugate}$

★ $x^n = -1$ $n = \text{even}$
 $2 \rightarrow \text{Real}(+1, -1)$
 $(n-2) \rightarrow \text{con \& conjugate}$

$$x^n = -1 \quad \left\{ \cos \frac{(2k+1)\pi}{n} + i \sin \frac{(2k+1)\pi}{n} \right\} \quad \{k=0, 1, 2, \dots, n-1\}$$

★ $\alpha \leftrightarrow \alpha^{n-1}$
 $\alpha^2 \leftrightarrow \alpha^{n-2}$
 $\alpha^3 \leftrightarrow \alpha^{n-3}$
 $\vdots$
 } conjugate pair chain



★ $i z \rightarrow \frac{\pi}{2}$ anti-clockwise
 $-i z \rightarrow \frac{\pi}{2}$ clockwise.

$$a+ib \rightarrow \pm \left\{ \sqrt{\frac{1}{2} [\sqrt{a^2+b^2} + a]} \pm i \sqrt{\frac{1}{2} [\sqrt{a^2+b^2} - a]} \right\}$$

$\begin{matrix} + \rightarrow +b \\ - \rightarrow -b \end{matrix}$

[B] Conjugate

(i) $[\bar{\bar{z}}] = z$

(ii) $z + \bar{z} = 2 \text{Re}(z)$

(iii) $z - \bar{z} = 2i \text{Im}(z)$

(iv) $z = \bar{z} \rightarrow z - \text{Real No}$

(v) $z = -\bar{z} \rightarrow z - \text{purely Imaginary No}$

$$(\bar{z}^n) = [\bar{z}]^n$$

$$\overline{z_1 + z_2 + z_3 + \dots + z_n} = \bar{z}_1 + \bar{z}_2 + \bar{z}_3 + \dots + \bar{z}_n$$

$$\overline{z_1 - z_2 - z_3 - \dots - z_n} = \bar{z}_1 - \bar{z}_2 - \bar{z}_3 - \dots - \bar{z}_n$$

$$\overline{z_1 z_2 z_3 z_4 z_5 \dots z_n} = \bar{z}_1 \cdot \bar{z}_2 \cdot \bar{z}_3 \cdot \bar{z}_4 \cdot \bar{z}_5 \dots \bar{z}_n$$

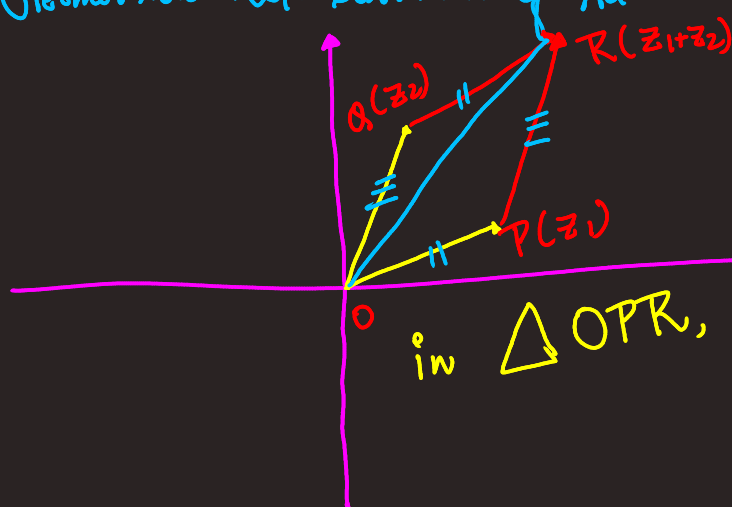
$$\left(\frac{\bar{z}_1}{\bar{z}_2} \right) = \frac{\bar{z}_1}{\bar{z}_2}$$

$\star z = |z| \{ \cos \theta + i \sin \theta \}$ Polar form
 $\star z = |z| \cdot e^{i\theta}$ Euler form

[C] Modulus Properties.

- [a] $z \cdot \bar{z} = |z|^2 = |\bar{z}|^2$
- [b] $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2)$
- [c] $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2 \operatorname{Re}(z_1 \bar{z}_2)$
- [d] $|z_1 - z_2|$ = distance b/w two Complex No of Argand plane
- [e] $|z_1 \cdot z_2 \cdot z_3 \cdot z_4 \dots z_n| = |z_1| \cdot |z_2| \cdot |z_3| \dots |z_n|$. A $(z_1) \leftrightarrow B(z_2)$
- [f] $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

★ Geometrical Representation of Addⁿ & Subⁿ of Complex No ★



$$|RP| = |z_1 + z_2 - z_1| = |z_2| = |RQ|$$

$$|RQ| = |z_1 + z_2 - z_2| = |z_1|$$

in $\triangle OPR$,

$$OP + PR > OR$$

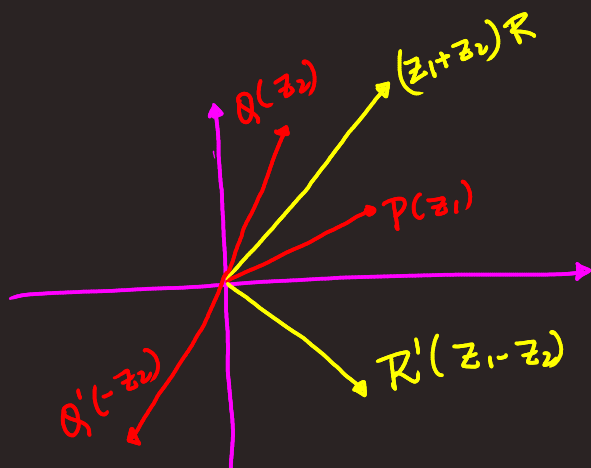
$$|z_1| + |z_2| > |z_1 + z_2|$$

$$|z_1| + |-z_2| > |z_1 - z_2|$$

if O, P, Q are in single line.

$$|z_1| + |z_2| = |z_1 + z_2| = |z_1 - z_2|$$

final $|z_1| + |z_2| > |z_1 \pm z_2|$ think if ask for Maxima
 think if ask Minima



in $\triangle OPR'Q'$

in $\triangle OPR'$, $|OR - PR'| < OR'$ Minima

Maxima $||z_1| - |z_2|| < |z_1 - z_2|$

$$||z_1| - |z_2|| < |z_1 + z_2|$$

if $z, 0, z_2$ are collinear points.

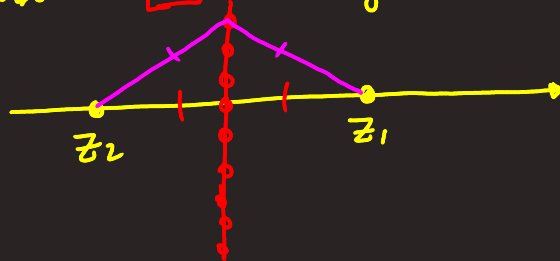
$$||z_1| - |z_2|| < |z_1 \pm z_2|$$

In general, $|z_1| + |z_2| + |z_3| + |z_4| + \dots + |z_n| > |z_1 + z_2 + z_3 + \dots + z_n|$

* Applⁿ of Modulus * Locus based on Distance Formula *

Let $A(z_1), B(z_2)$ are two fixed points. & z is Moving points

[a] $|z - z_1| = |z - z_2|$



[b] $|z - z_1| + |z - z_2| = |z_1 - z_2|$

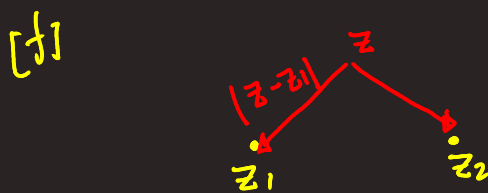
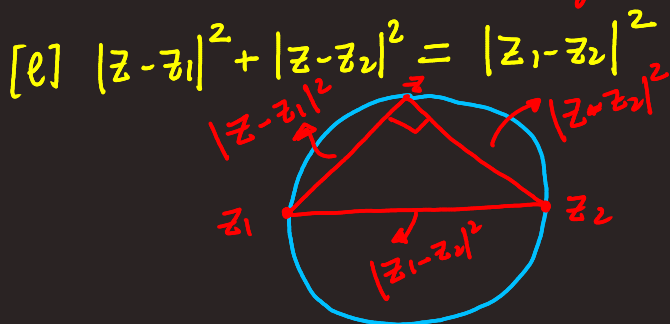


[c] $||z - z_1| - |z - z_2|| = |z_1 - z_2|$

2-Rays originates from z_1 & z_2



[d] $|z - z_1| = r$, z represent circle.
 $z_1 \rightarrow$ Co-ordinates of centre of circle.
 $r \rightarrow$ Radius of circle.



$z \rightarrow$ ellipse

$|z - z_1| + |z - z_2| = k$ if $k > |z_1 - z_2|$
 $|z_1 - z_2| = 2ae$
 $e = \frac{|z_1 - z_2|}{k}$



$z \rightarrow$ Hyperbola

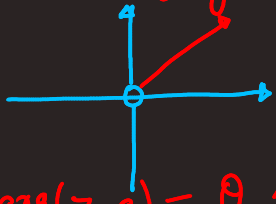
$||z - z_1| - |z - z_2|| = k$ if $k < |z_1 - z_2|$
 $|z_1 - z_2| = 2ae$
 $e = \frac{|z_1 - z_2|}{k}$

[h] $\frac{|z - z_1|}{|z - z_2|} = k$, if $k \neq 0, 1$

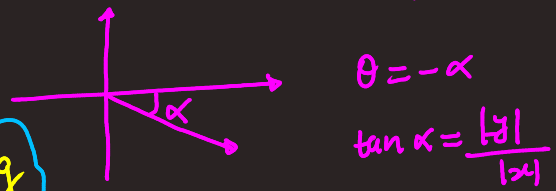
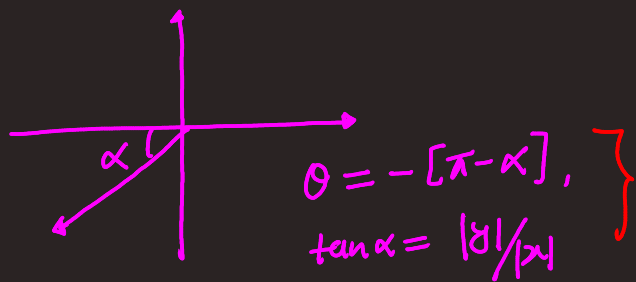
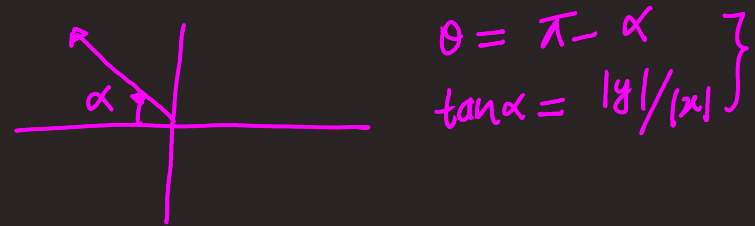
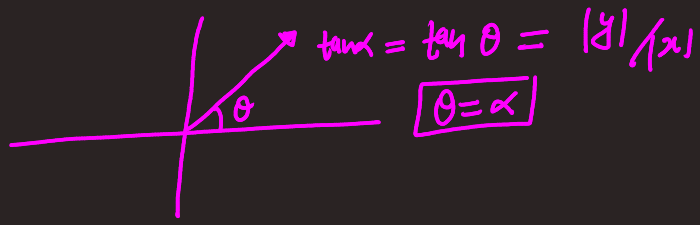
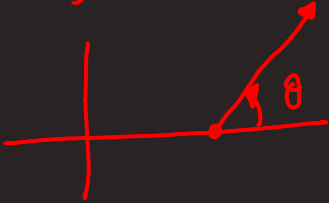
$z \rightarrow$ circle

[0] Argument. $\{-\pi < \theta < \pi\}$

* $\arg(z) = \theta, \{const\}$
 {ray excluding origin}



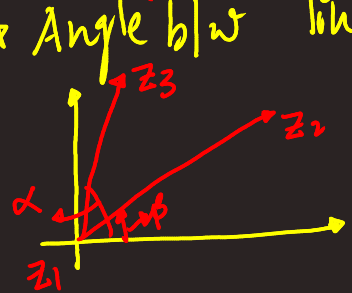
* $\arg(z-a) = \theta, \{const\}$ q70



* Properties of Argument * similar to log

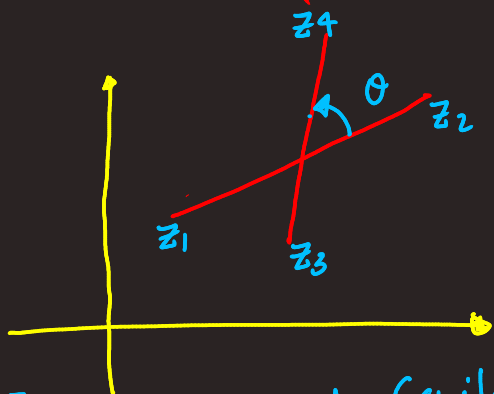
- [a] $\arg(\bar{z}) = -\arg(z)$
- [b] $\arg\left(\frac{1}{z}\right) = -\arg(z)$
- [c] $\arg(z) = 0, \pi$ $z \rightarrow \text{Real No}$
- [d] $\arg(z) = \frac{\pi}{2}, -\frac{\pi}{2}$ $z \rightarrow \text{Complex No}$
- [e] $\arg(z_1 \cdot z_2 \cdot z_3 \dots z_n) = \arg(z_1) + \arg(z_2) + \dots + \arg(z_n)$
- [f] $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

* Angle b/w lines (two)

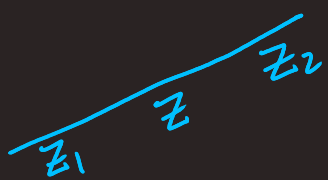


$$\alpha - \beta = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$$

Anti-clockwise, (+)
 Clockwise, (-)
 $\theta = \arg\left(\frac{z_4 - z_3}{z_2 - z_1}\right)$



* Eqn of lines $\{z_1, z_2\}$



* Equilateral $\Delta \{z_1, z_2, z_3\}$

★ Eqn of Arc ★

$z_1, z_2 \rightarrow$ two fixed Points & $z \rightarrow$ Variable Points

$$\arg \left(\frac{z-z_1}{z-z_2} \right) = \alpha \{ \text{fixed} \}$$

if $\alpha > 90^\circ$ Minor Arc.

if $\alpha < 90^\circ$ Major Arc

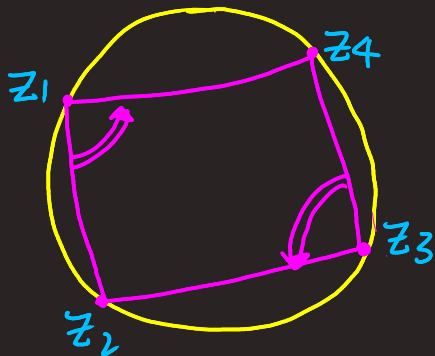


$$\star \arg \left(\frac{\text{final}}{\text{Initial}} \right)$$



$\rightarrow$ for check $(+,-)$ angle.

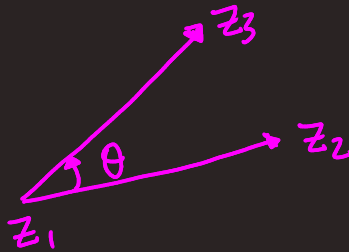
★ 4 points are be in Concyclic ★



$$\arg \left[\frac{z_1-z_4}{z_1-z_2} \right] + \arg \left[\frac{z_3-z_2}{z_3-z_4} \right] = \pi$$

★ Rotation Formula ★

$$\frac{\text{final}}{\text{Initial}}$$



$$\frac{z_3-z_1}{z_2-z_1} = \frac{|z_3-z_1|}{|z_2-z_1|} e^{i\theta} \text{ anti-clockwise.}$$

★ Binomial theorem ★ Chal = (8)

$$[x+y]^n = (n+1) \text{ terms}, \quad \sum_{r=0}^n {}^n C_r x^{n-r} y^r \quad \left\{ T_{r+1} = {}^n C_r x^{n-r} y^r \right\}$$

$$[x+y]^n = \sum_{r=0}^n T_{r+1}$$

★ rth term from end $T_{(n-r+1)+1}$

$$\begin{aligned} [x-y]^n &= \sum_{r=0}^n {}^n C_r x^{n-r} (-y)^r \\ &= \sum_{r=0}^n (-1)^r {}^n C_r x^{n-r} y^r \end{aligned}$$

$$[1+x]^n = \sum_{r=0}^n {}^n C_r x^r$$

$$[1-x]^n = \sum_{r=0}^n (-1)^r {}^n C_r x^r$$

★ Conjugate Expansion.

$$[x+y]^n + [x-y]^n = 2 [{}^n C_0 x^n y^0 + {}^n C_2 x^{n-2} y^2 + \dots]$$

$$[x+y]^n - [x-y]^n = 2 [{}^n C_1 x^{n-1} y^1 + {}^n C_3 x^{n-3} y^3 + \dots]$$

★ Middle term ★ $[x+y]^n$

[a] n = Even, only ONE Middle term,

[b] n = odd, Two middle term,

$\left\{ \frac{n+1}{2}, \frac{n+3}{2} \right\}$ th term

$$\begin{aligned} T_{\frac{n+1}{2}} &= T_{\frac{n-1}{2} + 1} \\ &= {}^n C_{\frac{n-1}{2}} \end{aligned}$$

$$\begin{aligned} \left[\frac{n}{2} + 1 \right] \text{th} \\ T_{\frac{n}{2} + 1} &= \\ {}^n C_{\frac{n}{2}} \end{aligned}$$

$$\boxed{r = \frac{n}{2}} \quad \begin{cases} n = 10, \\ (5) + 1 \end{cases}$$

$$\begin{aligned} T_{\frac{n+3}{2}} &= T_{\frac{n+1}{2} + 1} \\ &= {}^n C_{\frac{n+1}{2}} \end{aligned}$$

★ Multinomial Expansion ★

$$[x+y+z+w]^n = \sum \frac{n!}{\alpha! \beta! \gamma! \delta!} x^\alpha y^\beta z^\gamma w^\delta$$

$$\begin{aligned} \text{(a)} \quad 0 \leq \alpha, \beta, \gamma, \delta \leq n \\ \text{(b)} \quad \alpha + \beta + \gamma + \delta = n. \end{aligned}$$

★ No of distinct term = $n+r-1 (r-1)$

★ Application of Binomial theorem ★

[a] Approximation.

[b] Last some digit.

[c] Remainder theorem

[d] Divisibility check

$$[2+(\frac{1}{2})]^n = I + f \quad \left\{ \begin{array}{l} I = \text{integer} \\ f = \text{fractional part} \end{array} \right\}$$

★ Sum of Binomial Co-efficient ★

$$[x+y]^n = 2^n = {}^n C_0 + {}^n C_1 + {}^n C_2 + {}^n C_3 + \dots + {}^n C_n$$

$$[x=y=1] \quad 0 = {}^n C_0 - {}^n C_1 + {}^n C_2 - {}^n C_3 + \dots + {}^n C_n$$

$$[x=1, y=-1]$$

* $\{nCr = nC_{n-r}\}$

$nC_1 + nC_3 + nC_5 + \dots = \frac{2^n}{2} = nC_0 + nC_2 + nC_4 + nC_6 \dots$

★ Ratio of Consecutive Co-efficients / Terms in Binomial Expansion ★

$\frac{nCr}{nCr-1} = \frac{n - [r-1]}{r}$, Sum $\Rightarrow nCr + nCr-1 = \boxed{n+1Cr}$

★ Greatest Co-efficient in Binomial Expansion ★

$[a+bx]^n$, $r < \frac{n+1}{1 + |\frac{a}{bx}|}$ $\left| \frac{T_{r+1}}{T_r} \right| \gg 1, \left| \frac{T_{r+2}}{T_{r+1}} \right| < 1$

★ B.E for any Index ★

$[1+x]^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \frac{n(n-1)(n-2)(n-3)}{4!} x^4 + \dots \infty$

$T_r = \frac{n \cdot (n-1) \cdot (n-2) \dots (n-r+1)}{(r+1)!} x^{r+1}$

Condition, $n \notin \text{Natural}$, $|x| < 1$ infinite term
 $n = \text{Natural}$, $\rightarrow$ finite term $|x| < 1$ {Ma kses kahi krega}

[a] $[x+a]^n$
 $|x| > |a|$
 $x^n \left[1 + \frac{a}{x} \right]^n =$

[b] $[x+a]^n$
 $|x| < |a|$
 $a^n \left[1 + \frac{x}{a} \right]^n =$

★ Series involving in Binomial Co-efficient ★

[a] $\delta^{nCr} = \boxed{n \cdot n-1Cr-1}$
Same

[b] $\frac{nCr}{r+1} = \frac{n+1(r+1)}{n+1}$, **[c]** $\frac{nCr \cdot 2^r}{r+1} = \frac{n+1(r+1) \cdot 2^{r+1}}{2(n+1)}$

★ Series resulting from Patt of two Expansion ★
 [i] Upper index of both B.Co of Series Sum are fixed

- [a] Diff of lower indexes are constant
- [b] Sum of lower indexes are constant
- [ii] Upper indexes changes.

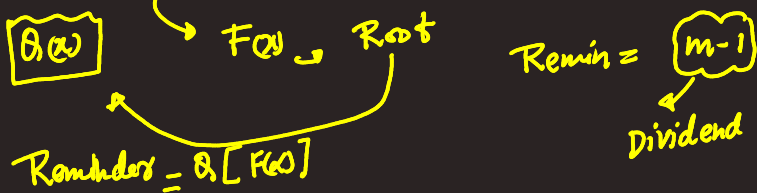
★ ★ Theory of Equation ★ ★

- (a) Real Polynomial $\rightarrow a_1, a_2, a_3, \dots, a_n \rightarrow \text{Real}$
- (b) Complex Polynomial $\rightarrow a_1, a_2, a_3, \dots, a_n \rightarrow \text{Complex}$
- (c) Rational Expression $\rightarrow \frac{p(x)}{q(x)} = 0 \Rightarrow p(x) = 0$
- (d) Deg of Poly $\rightarrow \begin{cases} \text{Complex} \\ \text{Real} \end{cases}$
- (e) Polynomial Eqn $\rightarrow \boxed{f(x) = 0}$

(f) Root of an eqn. value of variable

(g) Soln Set. Set of Root $\{a, b, c\}$

★ Remainder & Factor theorem ★



★ Identity & inequality $\left\{ \begin{array}{l} f(x) = ax^2 + bx + c \\ \text{Root} > \text{deg of } f(x) \end{array} \right\}$

$f(x) = ax^2 + bx + c$ $\xrightarrow{\alpha, \gamma}$ $f(x) = \text{Identity}$

$\boxed{a=b=c=0}$

★ Geometrical interpretation of zero of an expression and Root of an eqn ★ 2 graph

★ Domain of Eqn ★ $\frac{x^2 - 2x + 3}{x+1}$

★ Extraneous Roots ★ $\sqrt{x} = x-2$

★ Logs of root ★ $x^2 - 2x = x-2$

★ Graph of Polynomial Function ★

[a] $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots$

[b] $f(x) = (x-x_1)(x-x_2)(x-x_3) \dots$

} use Calcu $\begin{cases} x=0 \\ x \rightarrow \pm \infty \end{cases} y \rightarrow ??$

} easy, only see power of each bracket

Even $\rightarrow$ same sign

odd $\rightarrow$ opposite sign

★ ★ Equations reducible to quadratic ★ ★

[T-1] $ax^4 + bx^3 + cx^2 + dx + e = 0$ divided by x^2

[T-2] $(x-a)(x-b)(x-c)(x-d) = ex^2$

if $ac = bd$ divided by x^2

$ab = cd$

$ad = bc$

[T-3] $(x-a)^2 + (x-b)^2 = k$

$y = \frac{(x-a) + (x-b)}{2}$ $x = \text{cloud}$

[T-4] $[ax^2 + bx + c]^2 + [ax^2 + bx + d]^2 = k$

$\{t = ax^2 + bx\}$

★ Q.E & Nature of Roots ★

$y = ax^2 + bx + c$

(a) $a, b, c \rightarrow \text{Real}$ [many form]

(b) $a, b, c \rightarrow \text{complex}$ eqn

★ Range of Quadratic expression ★

$y = f(x)$ $y = \frac{f(x)}{g(x)}$ $\left\{ \begin{array}{l} \frac{x+b}{x+c} \\ D \neq 0 \end{array} \right\}$

★ Q.E in 2 variables ★★
 $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ $\begin{matrix} \text{L}_1 \\ \text{L}_2 \end{matrix}$
 $h^2 - ab > 0 \quad \Delta = 0$

★ Relation b/w root & Coefficient of Q.E ★★

(a) $\rightarrow \alpha + \beta = -\frac{b}{a}$

(b) transformation (a) symmetric form
 (b) Non-symmetric form

★ Common - Roots ★

$Q_1 = a_1x^2 + b_1x + c_1$

$Q_2 = a_2x^2 + b_2x + c_2$

Cond $\Rightarrow [a_1c_2 - a_2c_1]^2 = [a_1b_2 - a_2b_1][b_1c_2 - b_2c_1]$

(*) $\frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{c_1a_2 - a_1c_2} = \frac{1}{a_1b_2 - a_2b_1}$

(*) - Check (D) some time both roots common.

$\alpha \rightarrow$ Co-efft of x^2 (ii) - i

★ Q.F ★

$y = ax^2 + bx + c$

$y = a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} \right]$

$y = \left[a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a} \right]$

$y + \frac{D}{4a} = a \left(x + \frac{b}{2a} \right)^2$

★ Location of Root of Q.E ★

- 3 (0)
 a) +, + > 0
 b) -, - < 0
 c) +, -

- 3 (K)
 a) $> K$
 b) $< K$
 c) $\alpha > K > \beta$

- 3 (k_1, k_2)
 a) O.R b/w $\nearrow$
 b) both root b/w $\nearrow$
 c) $\alpha < k_1, k_2 < \beta$

★ Solving inequalities using location of roots ★

★ Newton Formula ★ $y = ax^2 + bx + c$ $\begin{matrix} \alpha \\ \beta \end{matrix}$

$S_n = \alpha^n + \beta^n$

$= aS_n + bS_{n-1} + cS_{n-2} = 0$